

Types of intents

Intents are divided into two groups:

- Explicit intents:** These explicitly specify the name of the target component to handle the intent; in these, the optional 'component name' field (above) is set to a particular value, through the `setComponent()` or `setClass()` methods.
- Implicit intents:** These do not specify a target component, but include enough information for the system to determine which of the available components is best to run for that intent. Consider an app that lists the available restaurants near you. When you click a particular restaurant option, the application has to ask another application to display the route to that restaurant. To achieve this, it could either send an explicit intent directly to the 'Google Maps' application, or send an *implicit intent*, which would be delivered to any application that provides the 'Maps' functionality (e.g., Yahoo Maps).

Intent Resolution

For implicit intents, Android 'figures out' the right component that can handle them. This process is called 'Intent Resolution', which maps an *Intent* to an Activity, Broadcast Receiver or Service (or sometimes two or more activities/receivers) that can handle it. If an application wants to receive a particular intent, it can do so in the two ways mentioned below:

- It should specify the same in the Android *Manifest.xml*, in the `<intent-filter>` tags, as shown below:

```
<intent-filter>
    <action android:name="android.intent.action.MAIN" />
    <category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
```

- It can explicitly register a Broadcast Receiver using `registerReceiver(BroadcastReceiver, IntentFilter)`

Types of intent broadcasts

There are three different types of broadcasts used by the Android System to broadcast intents:

- Normal Broadcast:** Sent asynchronously, all registered Broadcast Receivers will receive it. The order in which receivers receive it is undefined. Intents are sent with the `context.sendBroadcast()` method.
- Ordered Broadcast:** This is delivered sequentially to each eligible Broadcast Receiver; the order is defined by the priority of the associated Intent Filters, using the `android:priority` attribute in the Android *Manifest.xml*. Any Broadcast Receiver receiving an ordered broadcast can stop further sending by the `abortBroadcast()` method. The sender of an ordered broadcast can choose to be notified when the broadcast has completed. The `isOrderedBroadcast()` method returns true if the current receiver processes an ordered broadcast. These intents are sent using the `context.sendOrderedBroadcast()` method.

- Sticky Broadcast:** This is retained by the system even after it has been sent; this means that whenever you register a receiver for these intents, you will get a 'cached' copy of the intent that was recently sent. `ACTION_BATTERY_CHANGED`, an intent that describes the changes in battery status, is a good example of a sticky broadcast.

Sample code

The sample code available at http://linuxforu.com/article_source_code/dec12/intents_android.zip creates a simple intent to launch the contacts list; it also listens to the battery status change intent and provides the battery capacity. This sample code can be tested with Eclipse, running an Android emulator.

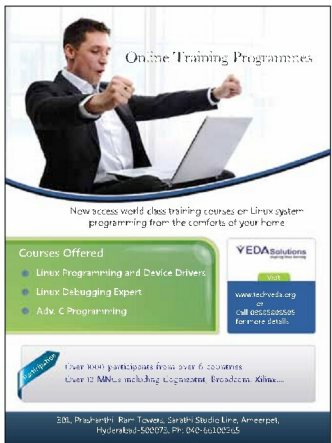
Having introduced readers to Android intents, in the next article, we will look at how Android prevents applications from sending system-level intents. 

References

- <http://developer.android.com/reference/android/content/Intent.html>

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